AERIS Expert Review Packet

Anomaly

Source / metric	tceq / no2
Observed / expected	27 / 2.6494
Time	2026-06-16 01:00 UTC
Location	29.6697, -95.1285
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	7 / 472	57.73 to 100; mean 88.7842	93.04 at 2026-06-16 00:55Z; dt 5 min
asos	temperature	degC	7 / 472	23 to 32.7778; mean 26.7684	26.7778 at 2026-06-16 00:55Z; dt 5 min
asos	wind_direction	degree	7 / 472	0 to 360; mean 120.339	80 at 2026-06-16 00:55Z; dt 5 min
asos	wind_speed	m/s	7 / 472	0 to 12.3467; mean 2.7837	2.57222 at 2026-06-16 00:55Z; dt 5 min
noaa_gfs	gh_500	m	8 / 96	5851.89 to 5926.57; mean 5887.01	5885.77 at 2026-06-16 00:00Z; dt 60 min
noaa_gfs	pbl_height	m	8 / 96	32.0322 to 1546.05; mean 357.83	48.199 at 2026-06-16 00:00Z; dt 60 min
noaa_gfs	precipitable_water	mm	8 / 96	53.4497 to 69.5435; mean 60.2728	60.9815 at 2026-06-16 00:00Z; dt 60 min
noaa_gfs	surface_pressure	hPa	8 / 96	1002.55 to 1014.43; mean 1009.3	1009.44 at 2026-06-16 00:00Z; dt 60 min
noaa_gfs	t_850	degC	8 / 96	16.785 to 19.498; mean 18.3156	18.8384 at 2026-06-16 00:00Z; dt 60 min
noaa_gfs	u_10m	m/s	8 / 96	-4.71304 to 1.58078; mean -0.9114	-0.619447 at 2026-06-16 00:00Z; dt 60 min
noaa_gfs	v_10m	m/s	8 / 96	-1.42555 to 6.14875; mean 2.0974	-0.575546 at 2026-06-16 00:00Z; dt 60 min
openaq	ozone	ppm	17 / 1110	-0.001 to 0.031; mean 0.0136	0.015 at 2026-06-16 01:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 212	5 to 110; mean 24.7783	20 at 2026-06-16 01:00Z; dt 0 min
openaq	pm25	ug/m3	87 / 3010	-2.6 to 52; mean 6.9224	9.1 at 2026-06-16 01:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 354	14 to 100; mean 92.8446	100 at 2026-06-16 01:01Z; dt 1.8 min
openweather	humidity	percent	5 / 354	68 to 100; mean 90.6497	95 at 2026-06-16 01:01Z; dt 1.8 min
openweather	precipitation	mm/h	5 / 354	0 to 24.39; mean 1.2444	0.24 at 2026-06-16 01:01Z; dt 1.8 min
openweather	pressure	hPa	5 / 354	1005 to 1016; mean 1011.31	1011 at 2026-06-16 01:01Z; dt 1.8 min
openweather	temperature	degC	5 / 354	24.18 to 33.3; mean 26.7748	26.57 at 2026-06-16 01:01Z; dt 1.8 min
openweather	wind_direction	degree	5 / 354	0 to 358; mean 159.61	92 at 2026-06-16 01:01Z; dt 1.8 min
openweather	wind_speed	m/s	5 / 354	0.07 to 5.67; mean 2.0357	0.7 at 2026-06-16 01:01Z; dt 1.8 min
purpleair	pm25	ug/m3	59 / 4307	0 to 46.2; mean 8.1124	6.4 at 2026-06-16 01:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 350.4 min
sentinel5p	s5p_ch4_granule_available	count	6 / 6	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 350.4 min
sentinel5p	s5p_co_column	mol/m^2	1 / 1	0.0259566 to 0.0259566; mean 0.026	0.0259566 at 2026-06-14 19:24Z; dt 1775.4 min
sentinel5p	s5p_co_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 350.4 min
sentinel5p	s5p_hcho_column	mol/m^2	1 / 1	0.000111617 to 0.000111617; mean 0.0001	0.000111617 at 2026-06-14 19:24Z; dt 1775.4 min
sentinel5p	s5p_hcho_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 350.4 min
sentinel5p	s5p_no2_column	mol/m^2	1 / 1	2.03572e-05 to 2.03572e-05; mean 0	2.03572e-05 at 2026-06-14 19:24Z; dt 1775.4 min
sentinel5p	s5p_no2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 350.4 min
sentinel5p	s5p_o3_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 350.4 min
sentinel5p	s5p_so2_column	mol/m^2	1 / 1	-6.10873e-06 to -6.10873e-06; mean -0	-6.10873e-06 at 2026-06-14 19:24Z; dt 1775.4 min
sentinel5p	s5p_so2_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; dt 350.4 min
tceq	co	ppm	3 / 213	0.1 to 1.1; mean 0.2221	0.2 at 2026-06-16 01:00Z; dt 0 min
tceq	no2	ppb	12 / 791	-2.1 to 27; mean 5.6245	27 at 2026-06-16 01:00Z; dt 0 min
tceq	so2	ppb	3 / 219	-0.6 to 0.6; mean -0.116	0.1 at 2026-06-16 01:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	06-14 12Z 79.64, 18Z 68.63, 06-15 00Z 86.88, 06Z 92.13, 12Z 92.06, 18Z 90.63, 06-16 00Z 92.98, 06Z 96.85, 12Z 92.15, 18Z 86.36, 06-17 00Z 89.64, 06Z 95.38, 12Z 93.95
asos	temperature	06-14 12Z 28.86, 18Z 30.89, 06-15 00Z 27.93, 06Z 27.03, 12Z 26.15, 18Z 26.04, 06-16 00Z 26.08, 06Z 25.52, 12Z 25.66, 18Z 26.96, 06-17 00Z 26.27, 06Z 24.77, 12Z 24.05
asos	wind_direction	06-14 12Z 150.3, 18Z 177.8, 06-15 00Z 125.6, 06Z 126.4, 12Z 168.3, 18Z 85.13, 06-16 00Z 69.75, 06Z 71.71, 12Z 164.2, 18Z 108.5, 06-17 00Z 104.6, 06Z 104.6, 12Z 114.3
asos	wind_speed	06-14 12Z 3.873, 18Z 4.27, 06-15 00Z 2.558, 06Z 2.587, 12Z 2.772, 18Z 1.134, 06-16 00Z 1.698, 06Z 1.117, 12Z 3.395, 18Z 2.083, 06-17 00Z 2.949, 06Z 4.191, 12Z 7.79
noaa_gfs	gh_500	06-14 18Z 5926, 06-15 00Z 5922, 06Z 5922, 12Z 5903, 18Z 5892, 06-16 00Z 5886, 06Z 5891, 12Z 5872, 18Z 5865, 06-17 00Z 5855, 06Z 5858, 12Z 5853
noaa_gfs	pbl_height	06-14 18Z 1403, 06-15 00Z 484.4, 06Z 225.2, 12Z 293.7, 18Z 372, 06-16 00Z 143.2, 06Z 72.41, 12Z 195.1, 18Z 413.1, 06-17 00Z 245.8, 06Z 131.1, 12Z 314.8
noaa_gfs	precipitable_water	06-14 18Z 59.07, 06-15 00Z 60.65, 06Z 59.96, 12Z 57.31, 18Z 64.94, 06-16 00Z 61.33, 06Z 62.99, 12Z 64.97, 18Z 63.2, 06-17 00Z 55.89, 06Z 56.78, 12Z 56.17
noaa_gfs	surface_pressure	06-14 18Z 1013, 06-15 00Z 1011, 06Z 1012, 12Z 1012, 18Z 1012, 06-16 00Z 1009, 06Z 1010, 12Z 1009, 18Z 1009, 06-17 00Z 1006, 06Z 1006, 12Z 1004
noaa_gfs	t_850	06-14 18Z 18.05, 06-15 00Z 19.1, 06Z 19.23, 12Z 18.69, 18Z 17.13, 06-16 00Z 18.77, 06Z 18.41, 12Z 17.75, 18Z 17.49, 06-17 00Z 19.03, 06Z 18.06, 12Z 18.09
noaa_gfs	u_10m	06-14 18Z -0.7691, 06-15 00Z -1.309, 06Z -0.3601, 12Z -0.3947, 18Z -1.215, 06-16 00Z -0.3869, 06Z -0.4979, 12Z 0.4683, 18Z 0.6401, 06-17 00Z -1.713, 06Z -2.308, 12Z -3.091

Source	Metric	Values
noaa_gfs	v_10m	06-14 18Z 4.444, 06-15 00Z 3.216, 06Z 2.513, 12Z 2.068, 18Z 3.56, 06-16 00Z -0.4505, 06Z 1.281, 12Z 2.212, 18Z 3.847, 06-17 00Z 0.4193, 06Z 1.134, 12Z 0.9245
openaq	ozone	06-14 12Z 0.01674, 18Z 0.02208, 06-15 00Z 0.01393, 06Z 0.01061, 12Z 0.008989, 18Z 0.01508, 06-16 00Z 0.007862, 06Z 0.002926, 12Z 0.01025, 18Z 0.01699, 06-17 00Z 0.01466, 06Z 0.02004, 12Z 0.01926
openaq	pm10	06-14 12Z 33.17, 18Z 22.17, 06-15 00Z 33.06, 06Z 49.44, 12Z 41.22, 18Z 12.22, 06-16 00Z 20.89, 06Z 23.69, 12Z 18.33, 18Z 15, 06-17 00Z 18.89, 06Z 14.44, 12Z 13.83
openaq	pm25	06-14 12Z 6.994, 18Z 8.435, 06-15 00Z 9.756, 06Z 9.691, 12Z 7.56, 18Z 4.627, 06-16 00Z 8.061, 06Z 7.203, 12Z 4.299, 18Z 2.702, 06-17 00Z 2.825, 06Z 4.83, 12Z 4.663
openweather	cloud_cover	06-14 12Z 56.6, 18Z 81.5, 06-15 00Z 95.83, 06Z 99.86, 12Z 99.13, 18Z 99.93, 06-16 00Z 99.57, 06Z 99.33, 12Z 99.9, 18Z 100, 06-17 00Z 96.81, 06Z 75.69, 12Z 77.6
openweather	humidity	06-14 12Z 83.15, 18Z 73.03, 06-15 00Z 89, 06Z 92.72, 12Z 93.48, 18Z 94.13, 06-16 00Z 92.93, 06Z 95.23, 12Z 94.17, 18Z 91.79, 06-17 00Z 91.1, 06Z 93.86, 12Z 95.4
openweather	precipitation	06-14 12Z 0.193, 18Z 0.1917, 06-15 00Z 0.203, 06Z 0.19, 12Z 3.307, 18Z 4.336, 06-16 00Z 0.7973, 06Z 1.585, 12Z 1.259, 18Z 0.7517, 06-17 00Z 0.4219, 06Z 0.7976, 12Z 3.892
openweather	pressure	06-14 12Z 1015, 18Z 1013, 06-15 00Z 1013, 06Z 1013, 12Z 1014, 18Z 1012, 06-16 00Z 1012, 06Z 1011, 12Z 1011, 18Z 1009, 06-17 00Z 1008, 06Z 1006, 12Z 1005
openweather	temperature	06-14 12Z 29.33, 18Z 31.85, 06-15 00Z 27.99, 06Z 27.09, 12Z 26.29, 18Z 25.43, 06-16 00Z 25.95, 06Z 25.36, 12Z 25.52, 18Z 26.41, 06-17 00Z 26.26, 06Z 25.04, 12Z 24.34
openweather	wind_direction	06-14 12Z 180, 18Z 182, 06-15 00Z 151.6, 06Z 164.7, 12Z 187.4, 18Z 163.7, 06-16 00Z 125.5, 06Z 198.5, 12Z 184.4, 18Z 165, 06-17 00Z 114.1, 06Z 111.2, 12Z 119.8
openweather	wind_speed	06-14 12Z 2.584, 18Z 3.247, 06-15 00Z 2.106, 06Z 2.097, 12Z 1.986, 18Z 1.815, 06-16 00Z 1.037, 06Z 1.571, 12Z 1.688, 18Z 2.229, 06-17 00Z 2.293, 06Z 2.014, 12Z 1.71
purpleair	pm25	06-14 12Z 8.091, 18Z 10.84, 06-15 00Z 11.51, 06Z 10.73, 12Z 8.115, 18Z 6.711, 06-16 00Z 9.756, 06Z 9.389, 12Z 4.485, 18Z 5.303, 06-17 00Z 4.857, 06Z 7.457, 12Z 8.394
sentinel5p	s5p_aer_ai_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_ch4_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_co_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_hcho_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_no2_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_o3_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	s5p_so2_granule_available	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
tceq	co	06-14 12Z 0.2067, 18Z 0.1933, 06-15 00Z 0.2167, 06Z 0.1778, 12Z 0.2333, 18Z 0.2667, 06-16 00Z 0.2778, 06Z 0.2278, 12Z 0.2556, 18Z 0.2444, 06-17 00Z 0.2167, 06Z 0.1556, 12Z 0.1833
tceq	no2	06-14 12Z 2.188, 18Z 1.518, 06-15 00Z 2.643, 06Z 2.819, 12Z 5.514, 18Z 7.59, 06-16 00Z 11.11, 06Z 8.731, 12Z 5.738, 18Z 7.23, 06-17 00Z 6.751, 06Z 4.889, 12Z 4.321
tceq	so2	06-14 12Z -0.24, 18Z -0.1278, 06-15 00Z -0.1833, 06Z -0.2389, 12Z -0.2111, 18Z -0.07222, 06-16 00Z 0.01667, 06Z -0.09444, 12Z -0.08333, 18Z -0.07222, 06-17 00Z 0.005556, 06Z -0.1, 12Z -0.15

Per-site averages

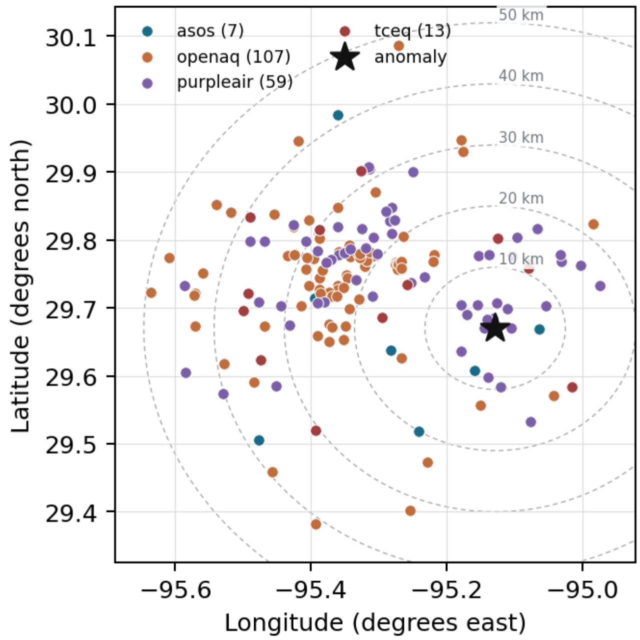
Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	T41 (6.225 km) 87.65, EFD (7.53 km) 87.67, HOU (15.295 km) 87.71, LVJ (20.026 km) 87.5, MCJ (26.207 km) 92.75, AXH (38.286 km) 86.26, IAH (41.543 km) 91.45

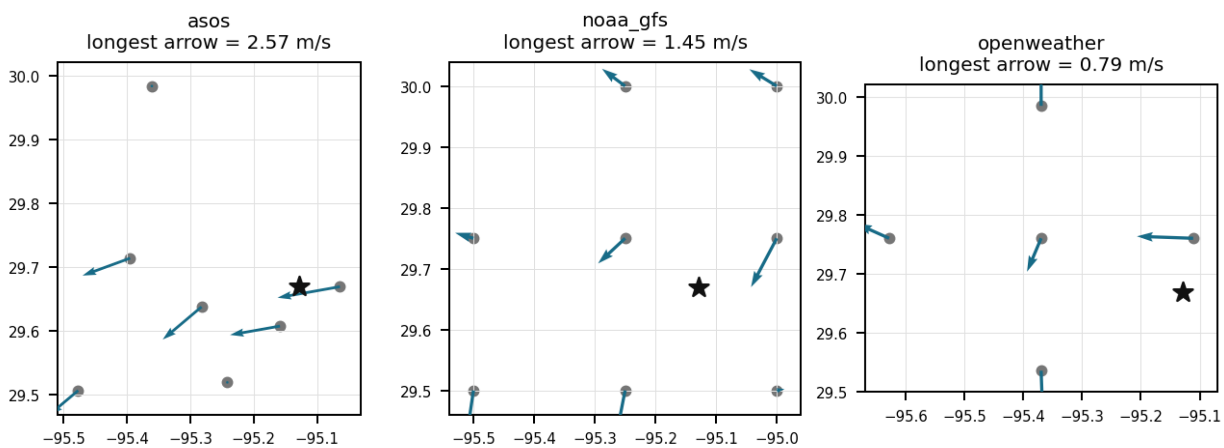
Source	Metric	Values
asos	temperature	T41 (6.225 km) 27.16, EFD (7.53 km) 27.15, HOU (15.295 km) 27.06, LVJ (20.026 km) 26.92, MCJ (26.207 km) 26.26, AXH (38.286 km) 26.57, IAH (41.543 km) 26.43
asos	wind_direction	T41 (6.225 km) 147.6, EFD (7.53 km) 105, HOU (15.295 km) 136.4, LVJ (20.026 km) 122.2, MCJ (26.207 km) 125.4, AXH (38.286 km) 63.89, IAH (41.543 km) 135
asos	wind_speed	T41 (6.225 km) 3.472, EFD (7.53 km) 3.46, HOU (15.295 km) 3.08, LVJ (20.026 km) 2.365, MCJ (26.207 km) 3.815, AXH (38.286 km) 1.15, IAH (41.543 km) 2.444
noaa_gfs	gh_500	gfs:29.75,-95.25 (14.741 km) 5887, gfs:29.75,-95.00 (15.288 km) 5886, gfs:29.50,-95.25 (22.231 km) 5888, gfs:29.50,-95.00 (22.598 km) 5887, gfs:29.75,-95.50 (36.97 km) 5887, gfs:30.00,-95.25 (38.548 km) 5887, gfs:30.00,-95.00 (38.76 km) 5887, gfs:29.50,-95.50 (40.577 km) 5888
noaa_gfs	pbl_height	gfs:29.75,-95.25 (14.741 km) 359.3, gfs:29.75,-95.00 (15.288 km) 359.7, gfs:29.50,-95.25 (22.231 km) 339.8, gfs:29.50,-95.00 (22.598 km) 438.5, gfs:29.75,-95.50 (36.97 km) 380.9, gfs:30.00,-95.25 (38.548 km) 369.1, gfs:30.00,-95.00 (38.76 km) 309.4, gfs:29.50,-95.50 (40.577 km) 305.9
noaa_gfs	precipitable_water	gfs:29.75,-95.25 (14.741 km) 60.6, gfs:29.75,-95.00 (15.288 km) 60.66, gfs:29.50,-95.25 (22.231 km) 60.64, gfs:29.50,-95.00 (22.598 km) 60.36, gfs:29.75,-95.50 (36.97 km) 59.84, gfs:30.00,-95.25 (38.548 km) 59.87, gfs:30.00,-95.00 (38.76 km) 59.73, gfs:29.50,-95.50 (40.577 km) 60.48
noaa_gfs	surface_pressure	gfs:29.75,-95.25 (14.741 km) 1009, gfs:29.75,-95.00 (15.288 km) 1010, gfs:29.50,-95.25 (22.231 km) 1010, gfs:29.50,-95.00 (22.598 km) 1011, gfs:29.75,-95.50 (36.97 km) 1008, gfs:30.00,-95.25 (38.548 km) 1008, gfs:30.00,-95.00 (38.76 km) 1009, gfs:29.50,-95.50 (40.577 km) 1009
noaa_gfs	t_850	gfs:29.75,-95.25 (14.741 km) 18.28, gfs:29.75,-95.00 (15.288 km) 18.3, gfs:29.50,-95.25 (22.231 km) 18.36, gfs:29.50,-95.00 (22.598 km) 18.48, gfs:29.75,-95.50 (36.97 km) 18.25, gfs:30.00,-95.25 (38.548 km) 18.27, gfs:30.00,-95.00 (38.76 km) 18.33, gfs:29.50,-95.50 (40.577 km) 18.26
noaa_gfs	u_10m	gfs:29.75,-95.25 (14.741 km) -0.7402, gfs:29.75,-95.00 (15.288 km) -0.7469, gfs:29.50,-95.25 (22.231 km) -1.209, gfs:29.50,-95.00 (22.598 km) -0.9102, gfs:29.75,-95.50 (36.97 km) -1.006, gfs:30.00,-95.25 (38.548 km) -0.6894, gfs:30.00,-95.00 (38.76 km) -0.7194, gfs:29.50,-95.50 (40.577 km) -1.27
noaa_gfs	v_10m	gfs:29.75,-95.25 (14.741 km) 1.617, gfs:29.75,-95.00 (15.288 km) 2.129, gfs:29.50,-95.25 (22.231 km) 2.181, gfs:29.50,-95.00 (22.598 km) 3.376, gfs:29.75,-95.50 (36.97 km) 1.62, gfs:30.00,-95.25 (38.548 km) 1.857, gfs:30.00,-95.00 (38.76 km) 2.167, gfs:29.50,-95.50 (40.577 km) 1.832
openaq	ozone	332 (0.032 km) 0.01806, 2800 (11.598 km) 0.01443, 2039 (14.096 km) 0.01161, 320 (14.272 km) 0.01255, 3378 (14.356 km) 0.009723, 5077791 (14.558 km) 0.01196, 3214 (14.765 km) 0.01543, 325 (16.163 km) 0.01329, +9 more
openaq	pm10	271 (0.032 km) 20.1, 13380082 (14.356 km) 27.03, 15852419 (21.358 km) 27.14
openaq	pm25	268 (0.032 km) 8.861, 13955815 (12.727 km) 6.365, 14404954 (13.725 km) 4.884, 2040 (14.096 km) 9.134, 14157975 (14.275 km) 8.057, 3379 (14.356 km) 8.533, 15539682 (14.359 km) 8.565, 8967123 (14.413 km) 9.868, +79 more
openweather	cloud_cover	grid:east (10.221 km) 93.38, grid:center (25.388 km) 94.24, grid:south (27.676 km) 93.3, grid:north (42.079 km) 88.85, grid:west (49.325 km) 94.48
openweather	humidity	grid:east (10.221 km) 90.7, grid:center (25.388 km) 91.9, grid:south (27.676 km) 88.21, grid:north (42.079 km) 91.32, grid:west (49.325 km) 91.13
openweather	precipitation	grid:east (10.221 km) 1.413, grid:center (25.388 km) 1.141, grid:south (27.676 km) 1.384, grid:north (42.079 km) 1.102, grid:west (49.325 km) 1.181
openweather	pressure	grid:east (10.221 km) 1011, grid:center (25.388 km) 1011, grid:south (27.676 km) 1011, grid:north (42.079 km) 1011, grid:west (49.325 km) 1011
openweather	temperature	grid:east (10.221 km) 27, grid:center (25.388 km) 26.84, grid:south (27.676 km) 26.88, grid:north (42.079 km) 26.62, grid:west (49.325 km) 26.54
openweather	wind_direction	grid:east (10.221 km) 165.8, grid:center (25.388 km) 157.9, grid:south (27.676 km) 154.9, grid:north (42.079 km) 161, grid:west (49.325 km) 158.4
openweather	wind_speed	grid:east (10.221 km) 2.601, grid:center (25.388 km) 1.556, grid:south (27.676 km) 2.518, grid:north (42.079 km) 2.011, grid:west (49.325 km) 1.486
purpleair	pm25	128007 (1.625 km) 2.899, 161689 (1.727 km) 8.023, 222589 (1.859 km) 8.111, 268165 (2.296 km) 8.786, 222593 (3.655 km) 7.803, 250205 (4.119 km) 6.425, 98633 (4.643 km) 4.663, 244939 (4.684 km) 8.312, +51 more
tceq	co	482011039 (0.0 km) 0.1319, 482011035 (14.355 km) 0.1352, 482011052 (29.756 km) 0.4029

Source	Metric	Values
tceq	no2	482011039 (0.0 km) 5.093, 482011015 (10.998 km) 4.789, 482011035 (14.355 km) 8.696, 482011050 (14.566 km) 1.444, 482010026 (14.787 km) 7.178, 482010416 (16.162 km) 5.649, 482011052 (29.756 km) 14.97, 480391004 (30.447 km) 2.258, +4 more
tceq	so2	482011039 (0.0 km) 0.02192, 482011035 (14.355 km) -0.4164, 482010051 (33.805 km) 0.04658

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

The PM2.5 levels are elevated at 6.4 ug/m3, which is higher than the mean of 8.1124 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

The increase in PM2.5 levels is due to an industrial accident at a nearby facility.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

The anomaly occurred at 2026-06-16T01:00:00+00:00 at the monitor located at 29.670, -95.129 in the Houston area.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

Air-quality co-pollutants near the anomaly are most consistent with a fresh local NOx plume rather than a regional secondary pollution event because ozone was low while PM2.5, PM10, CO, and SO2 were not simultaneously elevated to similarly extreme levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

A fresh local combustion plume from nearby mobile, industrial, marine, rail, or terminal activity is a likely physical cause of the 27 ppb NO2 spike because the anomaly occurred at one monitor with simultaneously low ozone, near-baseline CO, and no corresponding particulate spike, which is consistent with a localized NOx-rich plume rather than a broad regional pollution episode.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

At 2026-06-16T01:00:00+00:00 near the anomaly location, openaq reported ozone of 0.015 ppm, openaq reported PM2.5 of 9.1 ug/m3, openaq reported PM10 of 20.0 ug/m3, and tceq reported CO of 0.2 ppm and SO2 of 0.1 ppb.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

At the time of the NO2 anomaly on 2026-06-16 at 01:00:00 UTC, the nearest OpenAQ sensor recorded an ozone concentration of 0.015 ppm and a PM2.5 concentration of 9.1 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

Meteorological stagnation is supported because winds near the anomaly were weak at about 0.7 m/s in openweather and 2.57 m/s at the nearest asos station, while GFS planetary boundary layer height was only about 48 m near 2026-06-16 00:00 UTC, conditions favorable for trapping NO2 close to the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

A severe nitrogen dioxide concentration spike of 27.0 ppb was recorded by a TCEQ monitor at 29.670 N, -95.129 W on June 16, 2026, at 01:00 UTC, compared to an expected mean of 2.65 ppb.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 0.8455574567147295 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.16746984439502421 m/s; n=96 wind readings); openweather: event wind 0.7 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.4972179422298213 m/s; n=354 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

The air quality anomaly in Houston, Texas on June 16th, 2026 is characterized by elevated levels of PM2.5.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

The air quality anomaly in Houston, Texas on June 16th, 2026 is likely caused by a combination of factors including weather patterns and local emissions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

A local primary NOx plume is supported because the anomalous TCEQ monitor measured NO2 at 27.0 ppb at 2026-06-16 01:00 UTC while ozone at the co-located nearby OpenAQ sensor was only 0.015 ppm, a pattern consistent with fresh NOx emissions that suppress ozone near the source.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 0.8455574567147295 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.16746984439502421 m/s; n=96 wind readings); openweather: event wind 0.7 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.4972179422298213 m/s; n=354 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

OpenWeather cloud cover observations near 100 percent around 01:00 UTC on June 16, 2026, combined with nighttime photolysis suppression, allowed NO2 levels to remain elevated without photolytic loss to ozone.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

Regional TCEQ NO2 6-hour average concentrations peaked at 11.11 ppb during the 2026-06-16 00:00:00 UTC reporting interval.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

Meteorological conditions near eastern Houston around 2026-06-16 01:00 UTC favored trapping of primary emissions near the surface because boundary-layer height was very low, winds were weak, humidity was high, and cloud cover was nearly complete.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

A regional secondary or multi-pollutant pollution event is contradicted because particulate matter remained modest near the anomaly, with OpenAQ PM2.5 at 9.1 ug/m3, PurpleAir PM2.5 at 6.4 ug/m3, OpenAQ PM10 at 20 ug/m3, and TCEQ SO2 at only 0.1 ppb at the anomaly time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

OpenWeather cloud cover measurements exceeding 99% and low ambient ozone levels recorded by OpenAQ sensors indicate suppressed photolytic loss of NO2, promoting ground-level NO2 accumulation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

The anomaly is caused by a high-pressure system that has been lingering over the region.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

Industrial facility operations or heavy traffic corridors near the monitor served as the primary source of the elevated nitrogen dioxide emissions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

The air quality anomaly in Houston, Texas on June 16th, 2026 is also characterized by elevated levels of NO2.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

The TCEQ monitor at the anomaly location measured 27.0 ppb NO2 at 2026-06-16 01:00 UTC, far above the expected 2.65 ppb baseline, indicating an extreme local hourly enhancement at that site.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

A shallow planetary boundary layer and calm surface winds trapped localized combustion emissions near the surface in Houston around 01:00 UTC on June 16, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

The anomaly occurred on June 16th at approximately 1:00 AM, with the nearest sensor located about 1.625 km away.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

ASOS and OpenWeather meteorological observations recorded surface wind speeds dropping below 2 meters per second around 01:00 UTC on June 16, 2026, significantly restricting horizontal pollutant dispersion.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

Cloudy, humid nighttime conditions are a likely contributing cause of the high NO2 concentration because heavy cloud cover and low ozone indicate suppressed photolysis and weak oxidant production, which favors persistence of freshly emitted NO2 close to source regions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

Shallow nocturnal boundary-layer conditions and weak surface winds are a likely physical cause of the elevated NO_2 concentration because boundary-layer height near the event was very low and local winds were light, which would reduce dilution and allow near-surface NO_x to build up around the monitor.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

Low ambient ozone levels during the evening hours limited the chemical titration and photolytic destruction of nitrogen dioxide.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

NOAA GFS model output, ASOS surface observations, and OpenWeather meteorological data indicated a shallow planetary boundary layer height near 48 meters and light surface wind speeds below 2.6 m/s around the time of the NO_2 anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

A severe NO2 anomaly occurred at a TCEQ monitoring station at coordinates (29.670 N, -95.129 W) on 2026-06-16 at 01:00:00 UTC, reaching a concentration of 27.0 ppb compared to an expected value of 2.65 ppb.

Calm-wind flag: wind direction unstable under calm conditions. noaa_gfs: event wind 0.8455574567147295 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave 0.16746984439502421 m/s; n=96 wind readings); openweather: event wind 0.7 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.4972179422298213 m/s; n=354 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

The tceq NO2 monitor at 29.670, -95.129 recorded 27.0 ppb at 2026-06-16T01:00:00+00:00, compared with an expected value of 2.6493975903614455 ppb, corresponding to a z-score of 13.182458227775134 and severe anomaly classification.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

A combination of stagnant weather patterns and increased emissions from vehicles and industrial activities has contributed to the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

NOAA GFS model data indicates that the planetary boundary layer height collapsed to 143.2 meters around 00:00 UTC on June 16, 2026, establishing shallow vertical mixing that trapped surface NO_2 emissions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.